Customer Engagement Analysis

SECTION 1

What Is Customer Engagement?

Customer engagement measures how customers interact with the business over time. It goes beyond simple transaction counts to understand how often customers return to shop (frequency), how recently they last shopped (recency), how long they stay active (retention) and what patterns exist in their shopping behavior.

+88%	84-86%	94.5%	25 / yr
Customer base 6,046 to 11,350	Next month return rate	Repeat customers	Median visits per customer

Problem with aggregate metrics only

Total sales revenue doesn't tell the full story. A store could show any of these patterns.

- Growing revenue but losing customers (fewer customers spending more)
- Flat revenue but gaining customers (more customers spending less)
- The same customer count but declining engagement (customers shopping less frequently)

Key definitions

- **New Customer:** a customer making their very first purchase with the loyalty program
- **Repeat Customer:** a customer who has shopped before and returned within the last 3 months
- **Reactivated Customer:** a customer who returned after being inactive for 3+ months (a win-back)
- **Next Month Return Rate:** the percentage of customers active in month M who also shop in month M+1. This is the primary retention metric.

Note on retention
You might see very high return rates (80%+) in this dataset as it is a sample from actual business data. Real-world retail typically sees 30-50% next-month return rates.

Customer focus
This analysis focuses on loyalty program members only (is_loyalty = true).

SECTION 2

Monthly Engagement Breakdown

Customer base growth

- **Strong expansion:** the customer base grew from 6,046 in January 2025 to 11,350 in May 2026
- This represents **88% growth over 17 months**
- Growth was fastest in the first year (Jan-Dec 2025: 61% growth), then moderated in 2026 (16% growth)

Acquisition (new customers)

- January 2025 started at 100% new customers (program launch - all customers are new)
- By February 2025, new customers dropped to 17.3% as repeat behavior kicked in
- Stabilized at 4-5% from November 2025 onwards
- This low steady state is normal for mature loyalty programs - most growth comes from retention

Retention (repeat customers)

- Repeat customers grew from 0% to 94.5%+ of the active base
- This **94.5% repeat rate is excellent** and shows strong program stickiness
- Healthy businesses typically see 60-80% repeat customers; this exceeds that benchmark
- The high repeat % indicates customers are finding value in returning

Win-back (reactivated customers)

- Reactivations started appearing in May 2025 (4 months after launch) - expected, as customers need time to lapse before they can be reactivated
- Currently 1.1% of monthly active customers
- A low percentage is actually positive - it means few customers lapse long enough to need win-back

Retention health: next month return rate

- Return rates have been **remarkably stable at 84-86%** throughout the entire period
- No significant deterioration over time - indicates healthy, sustainable engagement
- Real-world retail typically sees 30-50% monthly return rates, so this is artificially high (synthetic data)

What this means

- **For real analysis:** these return rates would be exceptional and indicate extremely strong customer loyalty
- **For learning:** the technique is correct; real-world data would show lower but still meaningful patterns
- **Key insight:** stability matters more than absolute level - consistent rates mean predictable business performance

Seasonality

- No strong seasonal patterns observed in this data
- Return rates remained consistent across all months
- Customer counts grew steadily without major peaks or troughs
- Real retail often shows festival and holiday spikes - this synthetic data is smoothed

SECTION 3

Department-Level Engagement Patterns

Department engagement in May 2026

Analyzing the latest month (May 2026) to see how each department drives customer behavior.

Department	Customers	Repeat	Return rate
Grocery & Staples	10,973	93.9%	83.3%
Dairy & Frozen	10,854	93.4%	82.1%
Beverages	10,680	93.4%	82.9%
Household	7,170	90.0%	73.5%
Personal Care	1,951	78.2%	62.0%
Fashion & Lifestyle	1,604	69.8%	54.2%
Electronics & Accessories	1,043	61.1%	47.6%

High-engagement departments: the core loyalty drivers

These departments have the highest customer counts, highest repeat rates (90%+) and strongest return rates (80%+). They are the backbone of customer loyalty.

Grocery & Staples

- Core loyalty driver - highest engagement across all metrics; customers shop Grocery regularly (necessity purchases)
- Low reactivation % is good - customers don't lapse

> **Strategic value**
> Protect and invest - this drives store traffic for cross-shopping.

Dairy & Frozen

- Similar pattern to Grocery - high frequency, high loyalty; a regular replenishment category (milk, yogurt, frozen essentials)

> **Strategic value**
> Pair with Grocery for basket building - customers buy both together.

Beverages

- High-frequency category (tea, coffee, soft drinks) with consistent engagement like Grocery and Dairy

> **Strategic value**
> Strong impulse purchase opportunity - good for checkout placement.

Moderate-engagement department: Household essentials

- Solid performer with good repeat rates, but a lower return rate than the top tier
- Higher reactivation % (4.3%) suggests customers shop periodically, not continuously
- Products like cleaning supplies and home care are bought when needed, not on every trip

> **Strategic value**
> Promote regularly to drive more frequent purchases.

Specialty categories: lower frequency, moderate loyalty

These departments serve specific customer needs with less frequent shopping patterns.

Personal Care

- Moderate engagement - customers shop less frequently than essentials
- High reactivation % (7.9%) indicates customers return sporadically
- Products last longer (shampoo, soap), so repurchase frequency is lower

> **Strategic value**
> Focus on basket attachment - promote when customers shop Grocery/Dairy.

Discretionary departments: acquisition-heavy, low retention

These are infrequent purchase categories with high new customer rates but low return rates.

Fashion & Lifestyle

- Occasional purchase category - customers shop when they need clothing or accessories
- High new customer % (15.0%) - an acquisition gateway for some customers
- High reactivation % (15.3%) - customers return after long gaps
- A 54% return rate is normal for fashion - not a monthly need

> **Strategic value**
> Festival and seasonal promotions, not an everyday traffic driver.

Electronics & Accessories

- Most infrequent category - customers shop when they need specific items
- Very high new customer % - many customers try Electronics once
- Very high reactivation % - long gaps between purchases
- The 47.6% return rate reflects a durables buying pattern (buy a phone, return months later)

> **Strategic value**
> Focus on basket size per visit, not frequency. Bundle small electronics at checkout.

SECTION 4

Customer Behavior Statistics

Now that we've seen who is shopping (new, repeat, reactivated), the next step is to understand how they shop.

Transaction frequency: how often customers shop

- **Median (typical) customer:** shops 25 times per year (roughly every 2 weeks)
- **Above-average (P75):** 36 visits/year (every 10 days)
- **High-frequency (P90):** 73 visits/year (every 5 days)
- **Super-engaged (P95):** 78 visits/year (nearly twice per week)
- **Maximum:** 93 visits (someone who shops nearly 2x per week all year)
- Most customers shop twice a month - typical for combined grocery and occasional shopping
- There is a **3x gap between median (25) and P90 (73) shoppers**; the top 10% of customers visit 3x more often than the median customer
- **Opportunity:** move median customers toward P75 behavior through engagement programs

Total spend: customer lifetime value

- **Median (typical) customer:** ₹32,304 per year
- **Above-average (P75):** ₹60,827/year
- **High-value (P90):** ₹92,131/year
- **VIP (P95):** ₹113,980/year
- **Maximum:** ₹281,861 (extraordinary spending)
- The typical customer spends ₹32K annually across all departments
- The top 10% spend ₹92K+ - nearly 3x the median; the top 5% (VIP) spend ₹114K+ and are the most valuable customers
- The top 10% likely drive 40-50% of total revenue (typical in retail)
- **Revenue concentration:** with 10,000 active customers, median customers (5,000) contribute about ₹160M and the top 10% (1,000) contribute ₹100M+. Protecting the top 10% is critical - their churn hurts disproportionately.

Average basket size: spend per visit

- **Median (typical) customer:** ₹1,069 per trip
- **Above-average (P75):** ₹1,954 per trip
- **High-basket (P90):** ₹3,556 per trip
- **VIP (P95):** ₹4,538 per trip
- Most customers spend ₹1,000-1,100 per visit
- The top 10% spend ₹3,500+ per trip - they are buying either more items or more expensive items
- High total spend comes from both frequency and basket size: some customers shop often with small baskets (convenience), others shop less often but load up (bulk or planned shopping)

Recency: days since last purchase

- **Median:** 8 days (very recent)
- **P75:** 22 days (active but not super-frequent)
- **P90:** 52 days (approaching 2 months - the at-risk zone)
- **P95:** 95 days (3+ months - lapsed customer territory)
- Half the customer base shopped within the last 8 days - excellent engagement
- 10% haven't shopped in 52+ days and need win-back campaigns; 5% are 95+ days inactive - close to churned
- **Action triggers:** 30+ days - send reminder/offer; 60+ days - urgent win-back campaign; 90+ days - at risk of permanent churn

Inter-purchase interval: time between visits

- **Median:** 10.8 days between purchases (shop every 1.5 weeks)
- **P75:** 16.2 days (every 2-3 weeks)
- **P90:** 37 days (monthly shoppers)
- **P95:** 46 days (6-7 weeks between visits)
- The typical customer has a 10-day shopping rhythm, which explains the median of 25 visits/year (365 / 10.8 is about 34, adjusted for tenure)
- Customers have natural shopping cycles - respect these patterns; P90+ customers shop monthly or less and are periodic or planned shoppers

SECTION 5

Department-Level Shopping Behavior

High-frequency, small-basket departments

These drive store traffic through frequent, low-value visits.

Dairy & Frozen

- **Highest frequency of any department**; customers buy 1-2 items at a time (milk, eggs) and shop every 1-2 weeks consistently

Strategic value
This is the traffic engine. Ensure competitive pricing and availability, and use it as a prime location for cross-merchandising (pair with Beverages, Grocery).

Beverages

- Second-highest frequency (every 2-3 weeks) with small baskets (tea, coffee, soft drinks)
- Strong impulse category - good for checkout and end-caps; complements Dairy shopping trips

Grocery & Staples

- Moderate frequency (every 2-3 weeks) with small-to-medium baskets
- Necessity purchases - rice, dal, oil, staples; customers shop when they run out

Strategic value
Bundle with Dairy/Beverages for bigger baskets.

Moderate-frequency, medium-basket departments

Household

- Moderate frequency - monthly to bi-weekly shopping, with small baskets (cleaning supplies, paper products)
- Customers shop when they need it, not on every trip

Opportunity
Increase frequency through reminders and promotions.

Personal Care

- Very infrequent shoppers in this category; many shop Personal Care only 1-2x per year
- The 52-day median recency shows most customers haven't been recently; products last long (shampoo, soap), so repurchase need is low

Strategic value
Not a traffic driver. Focus on basket attachment when customers do visit, and promote during high-traffic periods (pair with Grocery runs).

Low-frequency, high-basket departments

These are planned purchases with infrequent visits but high spend per trip.

Fashion & Lifestyle

- Occasional shopping - customers visit only 1-2x per year, buying clothing and accessories in batches
- The 78-day recency means most customers haven't shopped this department recently
- P90 customers (17 visits/year) are fashion enthusiasts - segment them

> **Strategic value**
> Not about frequency - focus on basket size. Use festival and seasonal promotions (Diwali, wedding season) and identify fashion enthusiasts (P75+) for targeted marketing.

Electronics & Accessories

- Most infrequent department - customers shop 1-2x per year, buying phones, accessories and appliances
- The 96-day median recency means most customers last shopped Electronics 3+ months ago
- **Big spenders exist:** P90 customers spend ₹21K+/year in Electronics

> **Strategic value**
> A durables category, so low frequency is natural. Focus on basket size maximization and accessories attachment, target tech enthusiasts (P75+) for new launches, and use it as an acquisition channel - 20% of new customers come through Electronics.

Frequency vs basket size matrix

Department	Frequency (Median)	Basket Size (Median)	Strategic Play
Dairy & Frozen	19 visits/year	₹79	Traffic driver - protect at all costs
Grocery	15 visits/year	₹186	Traffic driver - bundle with Dairy
Beverages	14 visits/year	₹171	Traffic driver - impulse opportunity
Household	11 visits/year	₹151	Increase frequency through promotions
Personal Care	2 visits/year	₹236	Basket attachment to high-frequency
Fashion	2 visits/year	₹831	Maximize basket, seasonal events
Electronics	2 visits/year	₹1,447	Maximize basket, target tech enthusiasts

Cross-shopping opportunities

- Customers shopping Dairy (19x/year) should be cross-promoted into Household (11x/year)
- Promote small electronics and personal care items at Grocery/Dairy checkout
- **Festival bundles:** Grocery + Fashion/Electronics

Segment-specific strategies

- **Frequent small-basket shoppers (high Dairy/Grocery frequency):** goal - increase basket size. Tactic: bundle offers, larger pack sizes, cross-category promotions.
- **Infrequent high-basket shoppers (Electronics/Fashion):** goal - increase frequency or maximize each visit. Tactic: new product alerts, seasonal campaigns, loyalty rewards for the next purchase.
- **Balanced shoppers (moderate across all):** goal - move toward higher engagement. Tactic: personalized offers based on shopping history.

SECTION 6

Summary: How to Use These Insights

RFM customer segmentation

- Data-driven thresholds derived from percentile distributions
- Six actionable customer segments with distinct strategies
- **Revenue concentration: the top 22.4% of customers drive 52.8% of revenue**
- **VIP (6.7%):** ₹134K average spend, 21.8% of revenue - protect at all costs
- **Frequent Shoppers (15.7%):** ₹81K average spend, 31.0% of revenue - nurture toward VIP
- **Occasional Buyers (70.4%):** ₹26K average spend, 45.0% of revenue - the core majority; focus on engagement

Practical applications for marketing teams

Segment-specific campaigns

- **VIP (961 customers, 21.8% revenue):** white-glove service, exclusive perks, priority access. Retention is critical - losing 10% of VIPs means a ₹13M revenue loss.
- **Frequent Shoppers (2,246 customers, 31.0% revenue):** loyalty rewards, points programs, member-only sales. Upgrade campaigns such as "You're ₹23K away from VIP status!" Focus on basket-building (currently ₹2,562 vs VIP's ₹4,052).
- **Bulk Buyers (249 customers, 1.7% revenue):** frequency campaigns, convenience offers, free delivery. Goal: move from 9 visits/year to 20+ visits/year.
- **New Customers (509 customers, 0.3% revenue):** onboarding campaigns, welcome discounts, habit-building offers. Critical: drive 2nd and 3rd purchases within 60 days.
- **Occasional Buyers (10,096 customers, 45.0% revenue):** engagement campaigns to shift them toward Frequent Shoppers. Opportunity: if 10% upgrade to Frequent, **gain +₹55M revenue**.
- **Lapsers (273 customers, 0.2% revenue):** win-back campaigns, aggressive discounts, "We miss you" offers. Focus on 90-180 day lapsers (recently lapsed), not the 250+ day group.

Department-level campaigns

- **Traffic drivers (Dairy, Grocery, Beverages):** competitive pricing, ensure availability, prime placement for cross-merchandising
- **Revenue drivers (Electronics, Fashion):** maximize basket size, seasonal promotions, target P75+ customers
- **Cross-shopping:** promote low-frequency departments (Personal Care, Electronics) at high-frequency checkouts (Dairy, Grocery)

Practical applications for executive leadership: key health metrics to monitor

- **Monthly return rate:** 84-86% (currently stable). Red flag: drops below 80% for 2+ consecutive months. Benchmark: retail typically 30-50%; this business is significantly above average.
- **VIP + Frequent Shoppers % of revenue:** 52.8% (currently). Target: maintain or grow to 55-60%. Action: track how many Occasional customers become Frequent each quarter.
- **New customer acquisition:** 4-6% of monthly active customers. Stable at 4.4-5.7% across recent months. Goal: maintain a steady inflow while improving conversion to Repeat.
- **Lapser rate:** 1.9% of customers (currently). Target: keep below 3%. Prevention: catch customers at 60-90 day recency, before they hit 180+.
- **Revenue concentration risk:** the top 6.7% (VIP) drive 21.8% of revenue. Healthy: not over-dependent on a few customers (vs luxury retail where the top 5% = 50%+ of revenue). Monitor: if VIP churn rate exceeds 5%/year, implement a retention program immediately.

Strategic priorities

- **Protect the top:** VIP + Frequent (22.4% of customers, 52.8% of revenue). Churn prevention matters more than acquisition cost for these segments; budget accordingly - 60% of loyalty spend on the top 25% of customers.
- **Grow the middle:** move Occasional to Frequent. If 10% of Occasional (1,000 customers) upgrade, **gain +₹55M revenue**. Focus on P50-P75 customers (₹32K-60K spend) - they're closest to the Frequent threshold.
- **Onboard the new:** convert first-time to repeat within 60 days. 509 new customers/month × 60% conversion = 305 new repeat customers/month, which sustains growth and offsets natural churn.
- **Cross-sell strategically:** use high frequency to drive low frequency. Dairy/Grocery shoppers (10K+ customers) can be promoted Household/Personal Care, increasing basket size without new customer acquisition.

SECTION 7

Next Steps: From Analysis to Action

Immediate actions (this month)

- **Export customer lists by segment** and share with marketing for campaign targeting
- **Set up dashboards to track:** monthly return rate by segment, segment migration (how many moved Occasional to Frequent?) and VIP/Frequent churn rate
- **Launch pilot campaigns:** VIP - exclusive shopping hour, 10% off the entire purchase; Frequent - double points for 2 weeks; Occasional (near Frequent threshold) - "Spend ₹10K more this quarter, unlock Frequent benefits"

Short-term (next quarter)

- **Refine thresholds** based on 3 months of data: did P95 stay at ₹114K or shift? Adjust the VIP threshold accordingly, and test different thresholds in A/B campaigns
- **Implement department-specific promotions:** Dairy/Grocery - weekly specials to maintain traffic; Electronics/Fashion - seasonal events (Diwali, wedding season)
- **Build onboarding flows for new customers:** welcome email series, first-purchase discount, category tutorials

Long-term (next year)

- **Predictive churn modeling:** use this RFM baseline to build ML models and predict which Frequent Shoppers are at risk of downgrading to Occasional
- **Lifetime value (LTV) analysis:** calculate LTV by acquisition cohort and optimize acquisition spend by channel (which channels bring VIP-potential customers?)
- **Cross-department optimization:** bundle analysis (which department pairs drive the highest basket?) and store layout optimization (place high-basket departments near high-frequency departments)

SECTION 8

Key Takeaways

What makes a successful loyalty program? This analysis shows three pillars of effective customer engagement.

Know your segments (RFM)

- Not all customers are equal - VIP is not the same as Occasional
- Use data (percentiles) to define segments, not gut feel
- Tailor strategies by segment

Understand behavior drivers (department analysis)

- Dairy/Grocery drive frequency (habitual shopping)
- Electronics/Fashion drive basket size (planned purchases)
- Cross-promote to capture both

Track leading indicators (engagement metrics)

- Next month return rate = retention health
- New customer % = growth engine
- Recency distribution = churn early warning

> **Your retail business is healthy**
> strong retention (84-86% monthly return), balanced revenue (not over-dependent on a few whales), a growing base (**88% growth over 17 months**) and clear segments with actionable strategies.

The path forward

- **Protect VIP** (6.7%, 21.8% revenue) - churn = disaster
- **Grow Frequent** (15.7%, 31.0% revenue) - nurture toward VIP
- **Engage Occasional** (70.4%, 45.0% revenue) - prevent lapsing, upgrade to Frequent
- **Cross-sell strategically** - use high-frequency departments to drive low-frequency baskets

SECTION 9

Note on Synthetic Data

This analysis uses synthetic data generated for educational purposes. Real-world patterns will vary:

- **Seasonal effects:** real data shows stronger seasonality (festivals, holidays)
- **Regional differences:** customer behavior varies by geography (metro vs rural)
- **Economic factors:** inflation, income changes and competitor actions impact trends
- **Product mix:** your actual department mix will differ

When applying this framework to real data

- Validate percentiles against industry benchmarks
- Adjust thresholds for your specific business context
- Test segment definitions with historical cohorts
- Iterate based on campaign performance